CCS1101

INTRODUCTION TO COMPUTING

Comprehensive Lecture Notes

Lectures 1

Lecture 1: Introduction to Computing

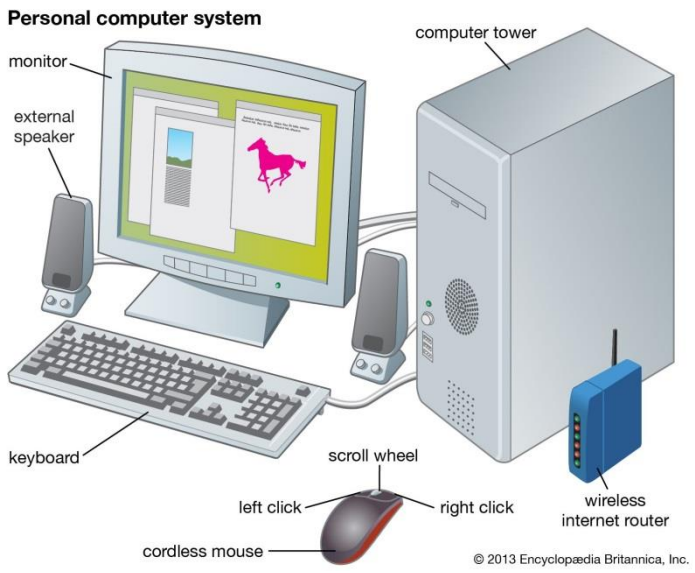
This opening lecture establishes the foundation for the entire course. It defines what a computer is, explores why computers matter to virtually every modern profession, and honestly examines both what computers can do exceptionally well and what they fundamentally cannot do.

1.1 Learning Objectives

- Define a computer and explain the relationship between data, information, and computation.
- State the key capabilities that make computers useful, and the key limitations that require human oversight.
- Describe the major generations of computing technology and how each transformed speed, size, and cost.
- Classify computers into categories based on size, purpose, and processing power.
- Explain, at a conceptual level, what it means to 'run a program.'

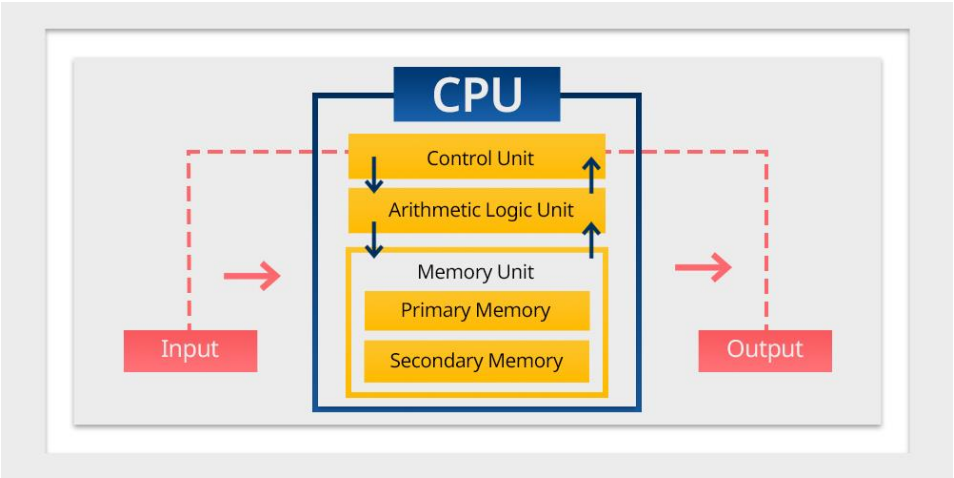
1.2 What Is a Computer?

Computer is an electronic device that processes data according to instructions provided by software programs. It takes input (data), processes it using a central processing unit (CPU), stores information, and produces output (results) to perform various tasks. It is widely used in fields such as education, business, communication, and entertainment for efficient and accurate operations.



Computing

Computing is the process of using computer hardware and software to manage, process, and transmit data. In simple terms, it is any activity that involves using a computer to solve problems, create content, or perform tasks.



1.3 Data, Information, and Computation

These three terms are often used loosely in everyday speech, but in computing they have precise, distinct meanings that this course will use consistently.

Term	Definition	Example
Data	Raw, unorganized facts with no context attached	The numbers 90, 85, 78, 92
Information	Data that has been organized, processed, or given context, making it meaningful	'The class average score is 86'
Computation	The process of transforming data into information by applying a defined procedure	Adding the four scores and dividing by 4

1.4 The Capabilities of Computers

Computers have become indispensable across nearly every field of human activity because of four defining capabilities:

1.4.1 Speed

Modern processors execute billions of instructions per second (a speed measured in GHz, or gigahertz — billions of cycles per second). A calculation that would take a human being years to complete by hand —

such as simulating a day of global weather patterns — can be completed by a supercomputer in minutes or hours.

1.4.2 Accuracy

When correctly programmed, a computer performs a given operation exactly the same way every single time, without the fatigue-induced errors, distraction, or inconsistency that affect human calculation. Errors that do occur are almost always traceable to incorrect instructions (a bug in the program) or incorrect input data, rather than the hardware 'making a mistake.'

1.4.3 Storage Capacity

A single modern hard drive can store trillions of characters of text, equivalent to millions of books, in a device that fits in the palm of a hand. This ability to store and instantly retrieve vast quantities of data underlies everything from search engines to medical records systems.

1.4.4 Automation and Repetition

Once programmed, a computer can repeat a task millions of times without complaint, without needing rest, and without any degradation in performance — a property essential to modern manufacturing, data processing, and scientific simulation.

1.5 The Limitations of Computers

Despite their remarkable capabilities, computers have fundamental limitations that every computing professional must understand and respect.

- **Lack of genuine understanding**: A computer manipulates symbols according to rules, but it does not 'understand' the meaning of those symbols the way a human does.
- **No common sense or independent judgment**: A computer will execute exactly what it is instructed to do, even if the instructions are flawed, nonsensical, or produce a harmful result. It cannot recognize on its own that an instruction 'doesn't make sense' unless it has been specifically programmed to check for that situation.
- **Complete dependency on instructions and data**: A computer cannot invent its own goals or decide what problem to solve; it requires a human-authored program and valid input data to do anything useful at all. This is often summarized by the classic phrase 'Garbage In, Garbage Out' (GIGO): if the input data or the instructions are flawed, the output will be flawed too, no matter how fast or precisely the computer executes them.
- **No creativity, emotion, or self-awareness (in the everyday human sense):** While computers can generate novel-looking output (such as AI-generated art or music), this output is fundamentally the result of applying learned statistical patterns, not genuine creative intent, emotional experience, or self-directed consciousness.

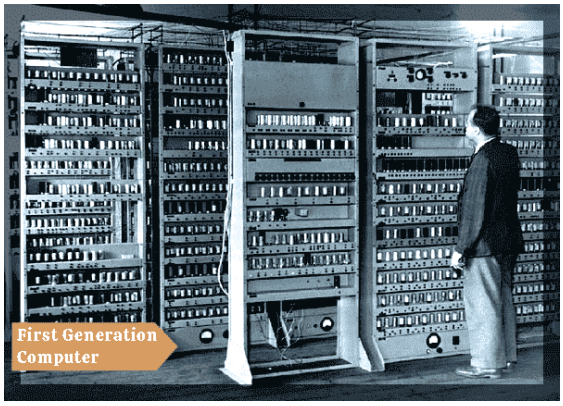
Example 1.1 — Capability vs. Limitation in Practice

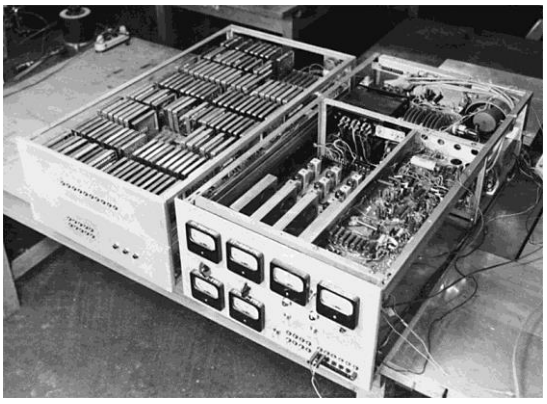
A hospital uses a computer system to calculate medication dosages based on a patient's weight and a standard formula. The computer performs this calculation with perfect accuracy and in a fraction of a second (a capability). However, if a nurse accidentally enters the patient's weight in the wrong unit (pounds instead of kilograms), the computer has no way of recognizing that something is medically implausible unless a human programmer specifically anticipated this error and added a validation check to the software (a limitation). This is precisely why critical systems always include human oversight alongside computational power.

1.6 A Brief History of Computing: The Five Generations

The evolution of computing technology is traditionally divided into five generations, each defined by a major advance in the underlying hardware technology.

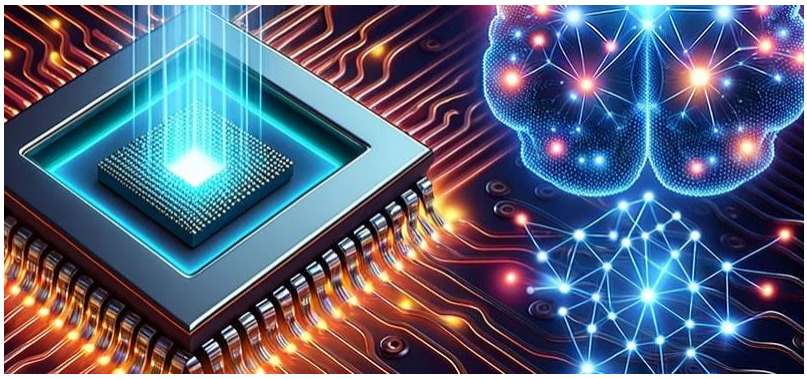
Generation	Approx. Period	Core Technology	Characteristics
First	1940s-1950s	Vacuum tubes	Enormous size, very high power consumption, extremely limited processing power
Second	1950s-1960s	Transistors	Smaller, faster, more reliable, and less power-hungry than vacuum tubes
Third	1960s-1970s	Integrated Circuits (ICs)	Multiple transistors on a single chip; introduction of operating systems
Fourth	1970s-present	Microprocessors	An entire CPU on a single chip; birth of the personal computer
Fifth	Present and emerging	Artificial Intelligence, parallel processing	Systems that learn, adapt, and process vast data in parallel









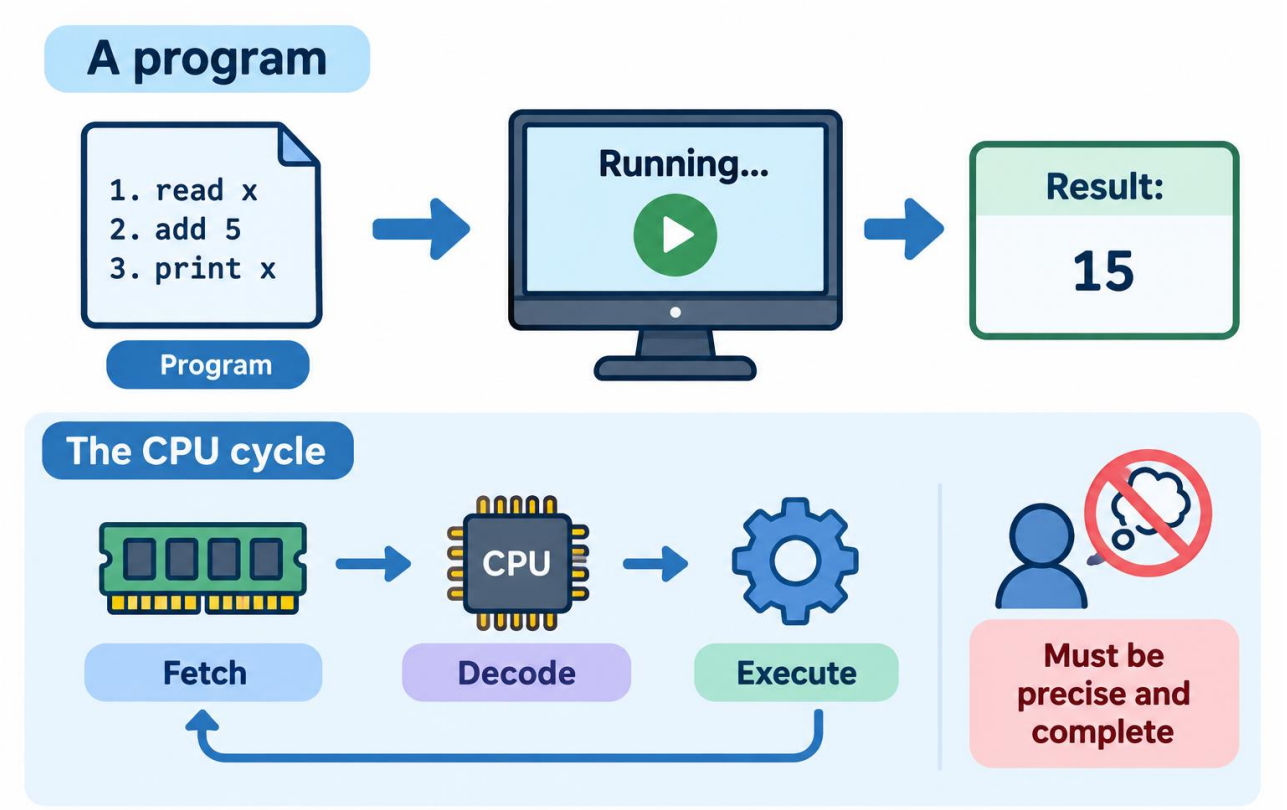


1.7 Types of Computers

Modern computers span an enormous range of sizes and purposes, from room-sized scientific instruments to devices embedded invisibly inside household appliances.

Type	Typical Use	Example
Supercomputer	Massive scientific simulations, weather forecasting, nuclear research	Frontier (Oak Ridge National Laboratory)
Mainframe	Large-scale transaction processing for banks, airlines, governments	IBM Z series
Server	Hosting websites, databases, and applications for many simultaneous users	Web/email/file servers
Personal Computer (Desktop/Laptop)	General personal and office productivity	Everyday desktop and laptop computers
Tablet / Smartphone	Portable, touch-based personal computing	Everyday mobile devices
Embedded System	A dedicated computer built into another device for one specific task	Car engine controllers, microwave ovens, smart thermostats

1.8 What Does It Mean to 'Run a Program'?



1.9 Pioneers of Computing

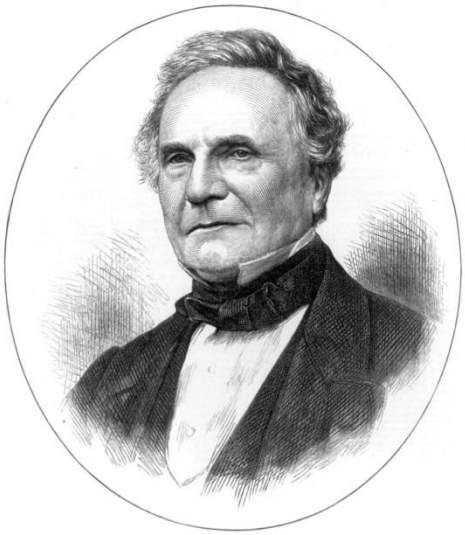
Understanding the people who shaped computing helps contextualize why the field is organized the way it is today. Three figures are especially foundational:

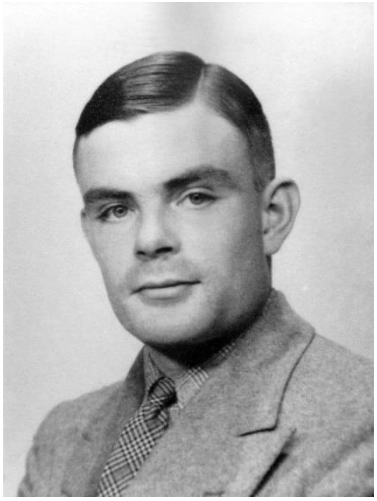
1.9.1 Charles Babbage (1791-1871)

Often called the 'father of the computer,' Babbage designed the Analytical Engine, a mechanical general-purpose computing machine that, although never fully built in his lifetime due to the engineering limitations of the era, incorporated concepts still central to computing today, including a processing unit, memory, and conditional branching (the ability to change what happens next based on a condition).

1.9.2 Alan Turing (1912-1954)

Turing formalized the theoretical foundation of computation itself through a mathematical model now called the Turing Machine, which defines, in the abstract, what it means for a problem to be 'computable' at all. Turing also played a pivotal role in codebreaking during the Second World War and later proposed the influential 'Turing Test' as a way of evaluating whether a machine could be said to exhibit intelligent behavior indistinguishable from a human's — an idea that remains a reference point in discussions of Artificial Intelligence.





1.10 Human vs. Computer: A Comparative Perspective

Task	Human Strength	Computer Strength
Arithmetic on large numbers	Slow, error-prone with fatigue	Extremely fast and perfectly accurate
Recognizing a familiar face in a crowd	Effortless, near-instant	Requires specialized training and large datasets (still imperfect)
Repeating an identical task 1,000,000 times	Tedious, error rate increases sharply	Trivial; no fatigue, no degradation
Understanding context, sarcasm, or ambiguity in language	Natural and intuitive	Difficult; still an active research challenge
Storing and instantly retrieving millions of records	Impossible without external aids	Routine and near-instantaneous

1.11 Review Questions

1. Define 'computer' using the four essential elements described in Section 1.2, and explain why the stored-program concept is considered so important.
2. Explain the difference between data and information, giving your own original example (not used in the lecture).
3. List two capabilities and two limitations of computers, and explain, using the hospital dosage example (or your own example), why human oversight remains necessary even with highly capable computer systems.
4. Which generation of computing introduced the microprocessor, and why was this development significant for the creation of the personal computer?
5. Classify each of the following as a supercomputer, mainframe, server, personal computer, or embedded system: (a) a bank's central transaction-processing machine, (b) a smart thermostat, (c) a national weather forecasting machine, (d) the laptop you use for coursework.
6. Briefly describe the contribution of one pioneer discussed in Section 1.9 (Babbage, Lovelace, or Turing), and explain why their contribution remains relevant to computing today.